

PHYS 301 Formula Sheet

Maxwell's Equations

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} & \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} & \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \\ \oint \mathbf{E} \cdot d\mathbf{a} &= \frac{Q_{\text{enc}}}{\epsilon_0} & \oint \mathbf{B} \cdot d\mathbf{a} &= 0 \\ \oint \mathbf{E} \cdot d\mathbf{l} &= -\frac{d\Phi_B}{dt} & \oint \mathbf{B} \cdot d\mathbf{l} &= \mu_0 I_{\text{enc}} + \epsilon_0 \mu_0 \frac{d\Phi_E}{dt}\end{aligned}$$

Electrostatics

Electric Field

Coulomb's law:

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}$$

Electric field:

$$\mathbf{F} = Q\mathbf{E} \quad \mathbf{E} = -\nabla V$$

E-field due to point charges:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i$$

E-field due to continuous charge distribution:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{\mathbf{r}}}{r^2} dq = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}') \hat{\mathbf{r}}}{r^2} d\tau'$$

Gauss's law:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

Electric Potential

$$V(b) - V(a) = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

Poisson's equation:

$$\nabla^2 V = -\rho/\epsilon_0$$

Potential due to continuous charge distribution:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r} dq = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r} d\tau'$$

Work and Energy in Electrostatics

Energy of point charges:

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i)$$

Energy of continuous charge distribution:

$$W = \frac{1}{2} \int \rho V d\tau \quad W = \frac{\epsilon_0}{2} \int_{\text{all space}} E^2 d\tau$$

Conductors

Electric field immediately outside a conductor:

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}$$

Surface charge:

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$$

Capacitors:

$$C = \epsilon_0 \frac{A}{d} \quad Q = CV$$

$$W = \frac{1}{2} CV^2 = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QV$$

Potentials

Laplace's equation:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

Separation of Variables

$$\int_0^a \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi x}{a}\right) dx = \begin{cases} 0 & \text{if } n \neq m \\ a/2 & \text{if } n = m \end{cases}$$

Legendre polynomials:

- $P_0(x) = 1$
- $P_1(x) = x$
- $P_2(x) = (3x^2 - 1)/2$
- $P_3(x) = (5x^3 - 3x)/2$

Solution to Laplace in spherical (ϕ independent):

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta)$$

Multipole Expansion

Axial multipole (α is between $\mathbf{r}$ and $\mathbf{r}'$):

$$\frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \alpha) \rho(\mathbf{r}') d\tau'$$

Monopole, Dipole, Quadrupole:

$$\begin{aligned}V_0(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \frac{Q}{r} & Q &= \int \rho(\mathbf{r}') d\tau' \\ V_1(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2} & \mathbf{p} &= \int \mathbf{r}' \rho(\mathbf{r}') d\tau' \\ V_2(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \sum_{ij} \frac{Q_{ij} \hat{r}_i \hat{r}_j}{r^3} \\ Q_{ij} &= \int \frac{\rho(\mathbf{r}')}{2} (3r'_i r'_j - (r')^2 \delta_{ij}) d\tau'\end{aligned}$$

Electric Fields in Matter

Dipole torque, force, and energy:

$$\mathbf{N} = \mathbf{p} \times \mathbf{E} \quad \mathbf{F} = \nabla(\mathbf{p} \cdot \mathbf{E}) \quad U = -\mathbf{p} \cdot \mathbf{E}$$

Bound charges:

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} \quad \rho_b = -\nabla \cdot \mathbf{P}$$

Electric displacement:

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad \oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}}$$

Linear Dielectrics

Polarization:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

Electric displacement:

$$\mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}$$

Energy:

$$W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

Boundary Conditions in Electrostatics

- $E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp} = \sigma/\epsilon_0$
- $E_{\text{above}}^{\parallel} = E_{\text{below}}^{\parallel}$
- $V_{\text{above}} = V_{\text{below}}$
- $D_{\text{above}}^{\perp} - D_{\text{below}}^{\perp} = \sigma_f$
- $D_{\text{above}}^{\parallel} - D_{\text{below}}^{\parallel} = P_{\text{above}}^{\parallel} - P_{\text{below}}^{\parallel}$

Magnetostatics

Lorentz force law:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Currents:

$$\mathbf{I} = \lambda \mathbf{v} \quad \mathbf{K} = \sigma \mathbf{v} \quad \mathbf{J} = \rho \mathbf{v}$$

Magnetic force on wire:

$$\mathbf{F}_{\text{mag}} = I \int (d\mathbf{l} \times \mathbf{B})$$

Continuity equation:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

Biot-Savart law:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\mathbf{r}}}{r^2} dl' = \frac{\mu_0}{4\pi} \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2}$$

Ampere's law:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

Magnetic Vector Potential

$\mathbf{B} = \nabla \times \mathbf{A}$ where $\nabla \cdot \mathbf{A} = 0$
Vector potential Poisson's equation:
 $\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$

Vector potential when $\mathbf{J} \rightarrow \mathbf{0}$ at infinity:
$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau'$$

Multipole expansion of a current loop:
$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \oint (r')^n P_n(\cos \alpha) d\mathbf{l}'$$

Magnetic dipole with vector area $\mathbf{a}$:
$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2}$$

$$\mathbf{m} = I \int d\mathbf{a} = I\mathbf{a}$$

Magnetic Fields in Matter

Dipole torque, force, and energy:
$$\mathbf{N} = \mathbf{m} \times \mathbf{B} \quad \mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B}) \quad U = -\mathbf{m} \cdot \mathbf{B}$$

Bound currents:
$$\mathbf{J}_B = \nabla \times \mathbf{M} \quad \mathbf{K}_B = \mathbf{M} \times \hat{\mathbf{n}}$$

Auxiliary field:
$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M} \quad \oint \mathbf{H} \cdot d\mathbf{l} = I_{f,\text{enc}}$$

Linear Media

Magnetization:
$$\mathbf{M} = \chi_m \mathbf{H}$$

Auxiliary field:
$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu_0(1 + \chi_m)\mathbf{H} = \mu\mathbf{H}$$

Boundary Conditions in Magnetostatics

- $B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0 K$
- $B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}$
- $\mathbf{A}_{\text{above}} = \mathbf{A}_{\text{below}}$
- $H_{\text{above}}^{\perp} - H_{\text{below}}^{\perp} = -(M_{\text{above}}^{\perp} - M_{\text{below}}^{\perp})$
- $H_{\text{above}}^{\parallel} - H_{\text{below}}^{\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}}$

Miscellaneous Formulas

Electric field of dipole:
$$\mathbf{E}_{\text{dip}}(r, \theta) = \frac{\mathbf{p}}{4\pi\epsilon_0 r^3} \left(2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}} \right)$$

Electrodynamics

Force per unit charge:
$$\mathbf{f} = \mathbf{v} \times \mathbf{B}$$

Electromotive force:
$$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} \quad \mathcal{E} = vBh$$

Flux rule:
$$\mathcal{E} = -\frac{d\Phi}{dt}$$

Vector Derivatives

Cartesian
Gradient:
$$\nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{x}} + \frac{\partial f}{\partial y} \hat{\mathbf{y}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}}$$

Divergence:
$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

Curl:
$$\nabla \times \mathbf{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}}$$

Spherical
Gradient:
$$\nabla f = \frac{\partial f}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\boldsymbol{\phi}}$$

Divergence:
$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_{\theta}) + \frac{1}{r \sin \theta} \frac{\partial v_{\phi}}{\partial \phi}$$

Curl:
$$\nabla \times \mathbf{v} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta v_{\phi}) - \frac{\partial v_{\theta}}{\partial \phi} \right] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_{\phi}) \right] \hat{\boldsymbol{\theta}} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r v_{\theta}) - \frac{\partial v_r}{\partial \theta} \right] \hat{\boldsymbol{\phi}}$$

Cylindrical
Gradient:
$$\nabla f = \frac{\partial f}{\partial s} \hat{\mathbf{s}} + \frac{1}{s} \frac{\partial f}{\partial \phi} \hat{\boldsymbol{\phi}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}}$$

Divergence:
$$\nabla \cdot \mathbf{v} = \frac{1}{s} \frac{\partial}{\partial s} (s v_s) + \frac{1}{s} \frac{\partial v_{\phi}}{\partial \phi} + \frac{\partial v_z}{\partial z}$$

Curl:
$$\nabla \times \mathbf{v} = \left[\frac{1}{s} \frac{\partial v_z}{\partial \phi} - \frac{\partial v_{\phi}}{\partial z} \right] \hat{\mathbf{s}} + \left[\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s} \right] \hat{\boldsymbol{\phi}} + \frac{1}{s} \left[\frac{\partial}{\partial s} (s v_{\phi}) - \frac{\partial v_s}{\partial \phi} \right] \hat{\mathbf{z}}$$

Fundamental Theorems

Gradient theorem:
$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla f) \cdot d\mathbf{l} = f(\mathbf{b}) - f(\mathbf{a})$$

Divergence theorem:
$$\int (\nabla \cdot A) d\tau = \oint \mathbf{A} \cdot d\mathbf{a}$$

Stoke's theorem:
$$\int (\nabla \times A) \cdot d\mathbf{a} = \oint \mathbf{A} \cdot d\mathbf{l}$$

Spherical Coordinates

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \\ \hat{\mathbf{x}} = \sin \theta \cos \phi \hat{\mathbf{r}} + \cos \theta \cos \phi \hat{\boldsymbol{\theta}} - \sin \phi \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{y}} = \sin \theta \sin \phi \hat{\mathbf{r}} + \cos \theta \sin \phi \hat{\boldsymbol{\theta}} + \cos \phi \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} = \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}} \\ r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \tan^{-1} \left(\sqrt{x^2 + y^2} / z \right) \\ \phi = \tan^{-1} (y/x) \\ \hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}} \\ \hat{\boldsymbol{\theta}} = \cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}} \\ \hat{\boldsymbol{\phi}} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}} \end{cases}$$

Cylindrical Coordinates

$$\begin{cases} x = s \cos \phi \\ y = s \sin \phi \\ z = z \\ \hat{\mathbf{x}} = \cos \phi \hat{\mathbf{s}} - \sin \phi \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{y}} = \sin \phi \hat{\mathbf{s}} + \cos \phi \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} = \hat{\mathbf{z}} \\ s = \sqrt{x^2 + y^2} \\ \phi = \tan^{-1} (y/x) \\ z = z \\ \hat{\mathbf{s}} = \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}} \\ \hat{\boldsymbol{\phi}} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}} \\ \hat{\mathbf{z}} = \hat{\mathbf{z}} \end{cases}$$